

06. Conclusion & Engineering Roadmap

This document summarizes the technical achievements of the **PredSea** oceanographic forecasting architecture and outlines the future engineering roadmap.

1. Summary of Architectural Accomplishments

PredSea demonstrates that high-resolution (1 km) regional ocean modeling does not require expensive, dedicated supercomputer infrastructure. By combining serverless GCP Batch Spot orchestration, parallel process-level Python pre-processing, and rigorously validated Fortran numerical engines (CROCO, SWAN, WRF), PredSea achieves:

1. **High-Resolution Coastal Granularity:** Resolves 1 km coastal channels, island wind shadowing, and bathymetric shoaling across the Balearic Sea and Western Mediterranean.
 2. **Order-of-Magnitude Acceleration:** Reduces pre-processing input assembly times from **40 minutes to 1 min 42 sec** using process-level GIL bypass techniques, and completes total regional ocean execution in **under 35 minutes**.
 3. **Cost Efficiency:** Operates the entire multi-region forecast suite for **~2.22 per day**, representing a **>90% cost reduction** compared to monolithic, always-on cloud instances.
 4. **Physical & Empirical Accuracy:** Verified against SOCIB buoy observations and satellite SST baselines (July 2026 Gate 8c mean SST 28.35° C), completely eliminating past 40° C thermal runaway bugs.
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2. Technical & Strategic Roadmap

Milestone A: Multi-Day Diurnal Thermal Skin Layer Tracking

While Gate 8c validates 24-hour heat flux equilibrium, multi-day forecasting (96 – 120 hours) during summer marine heatwaves requires higher vertical resolution in the upper 1 meter ocean skin layer:

- **s-Coordinate Refinement:** Increase vertical levels from $N = 32$ to $N = 40$ or $N = 50$, increasing surface stretching ($\theta_s = 8.0$) to place 5 vertical layers within the top 1 meter.
- **Diurnal Warm-Layer Modeling:** Integrate explicit Cool-Skin / Warm-Layer parameterizations (e.g. Fairall et al. COARE scheme) to track diurnal surface warming peaks (+1.5° C – 2.5° C afternoon spikes) and nocturnal mixing decay.

Milestone B: Full Operational CROCO-SWAN Wave-Current Coupling

Expand two-way standalone runs into active three-way dynamic OASIS3-MCT coupled cycles:

- **Wave Radiation Stress Feedback:** Dynamically pass SWAN S_{xx}, S_{xy}, S_{yy} radiation stress gradients into CROCO to drive wave-induced longshore currents, wave setup in harbors, and wave-current bottom friction enhancement.
- **Current Refraction Feedback:** Pass CROCO 1 km hourly surface currents back into SWAN to compute Doppler-shifted wave refraction in high-current channels (e.g., Ibiza-Formentera Freus channel).

Milestone C: Automated Captain-Facing Decision APIs & Alerting

Bridge raw numerical model outputs directly to operational maritime decision tools:

- **Vessel Response Threshold Engine:** Translate 1 km wave spectra, wind against current vectors, and cross-channel steepness into vessel-class safety statuses (favorable, workable,

conservative, restricted) for small (< 12 m), medium ($12 - 24$ m), and large (> 24 m) motor and sailing yachts.

- **Automated Artifact Dispatch:** Automatically compile daily briefing maps, WhatsApp captain advisories, and LinkedIn operational summaries upon forecast completion.
 - **FastAPI / Deck.gl Production Endpoints:** Serve real-time oceanographic vector fields and wave condition layers to mobile and web dashboards at sub-second latencies.
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3. Concluding Remarks

The PredSea White Paper establishes a validated blueprint for next-generation, cloud-native oceanographic forecasting. By combining numerical rigor in hydrodynamic physics with modern serverless cloud infrastructure, PredSea provides maritime operators, captains, and harbor authorities with accurate, reliable, and cost-effective decision intelligence for the sea.